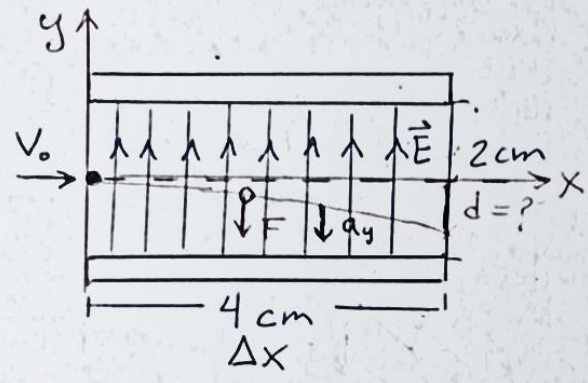


F1



En la dirección horizontal:

$$\Delta x = v_0 t$$

$$t = \frac{\Delta x}{v_0}$$

$$t = \frac{0,04 \text{ m}}{2 \times 10^7 \text{ m/s}}$$

$$t = 2 \times 10^{-9} \text{ (s)}$$

Como $F = q E$
 $F = e E$
 $\sum F_y = m a_y$
 $F = m a_y$

$$a_y = \frac{F}{m} = \frac{(1,602 \times 10^{-19})(200000)}{9,109 \times 10^{-31}}$$

$$a_y = 3,517 \times 10^{15} \text{ m/s}^2$$

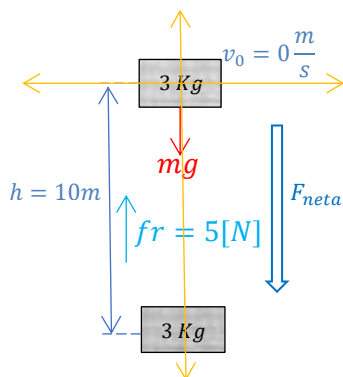
$$\therefore \Delta y = d = \cancel{v_0 t} + \frac{1}{2} a_y t^2$$

$$\Delta y = \frac{1}{2} (3,517 \times 10^{15}) (2 \times 10^{-9})^2$$

$$d = 7,03 \times 10^{-3} \text{ m}$$

$$d \cong 7,0 \text{ mm}$$

F2



$$F_{neta} = mg - fr$$

$$F_{neta} = 3[\text{kg}] * 10 \left[\frac{\text{m}}{\text{s}^2} \right] - 5[\text{N}]$$

$$F_{neta} = 25[\text{N}]$$

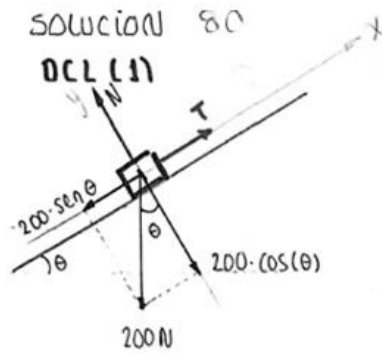
Como el desplazamiento es 10 m

$$W_{neto} = F_{neta} * \Delta y$$

$$W_{neto} = 25[\text{N}] * 10[\text{m}]$$

$$W_{neto} = 250[\text{Joul}]$$

F3



Para (1)

$$\sum F_x = 0$$

$$T - 200 \cdot \sin \theta = 0$$

$$100 - 200 \cdot \sin \theta = 0$$

$$-200 \cdot \sin \theta = -100$$

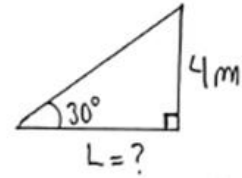
$$\sin \theta = \frac{100}{200}$$

$$\sin \theta = \frac{1}{2}$$

$$\theta = \sin^{-1}\left(\frac{1}{2}\right)$$

$$\boxed{\theta = 30^\circ}$$

Del triángulo



$$\tan(30^\circ) = \frac{4}{L}$$

$$L = \frac{4}{\tan(30^\circ)}$$

$$L = 6,928(m) \approx 7[m]$$

F4

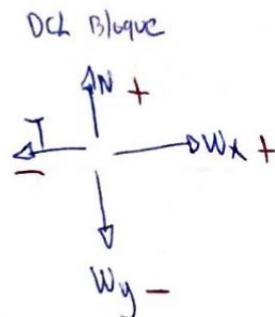
Datos

$$W = 3$$

$$q_A = 3 \times 10^6$$

$$q_B = -5 \times 10^6$$

$$W_B = 1.5 N$$



$$\sum F_y = 0$$

$$T - W_B - F = 0$$

$$T - 1.5 - 9 \times 10^9 \times \frac{3 \times 10^6 \times 5 \times 10^6}{0.3^2} = 0$$

$$T - 1.5 - 1.5 = 0$$

$$T = 3 \text{ IN}$$

$$\sum F_x = \text{max}$$

$$W_x - T = 0$$

$$W \sin 37^\circ - 3 = 0$$

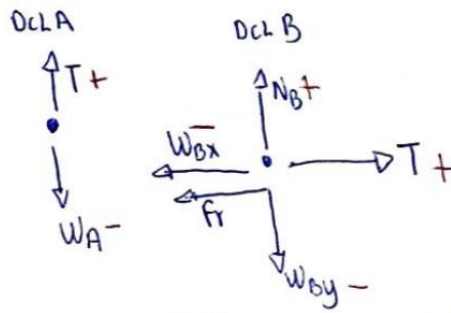
$$\boxed{W = 5 \text{ IN}}$$

F5

$$\mu = ?$$

$$W_A = 40 \text{ N}$$

$$W_B = 50 \text{ N}$$



$$\sum F_y = 0$$

$$T - W_A = 0$$

$$T = 40 \text{ IN}$$

$$\sum F_y = 0$$

$$N_B - W_{by} = 0$$

$$N_B - W_B \cos 37 = 0$$

$$N_B = 50 \cos 37$$

$$N_B = 39.93 \text{ IN}$$

$$\sum F_x = 0$$

$$T - W_{bx} - F_r = 0$$

$$40 - 50 \sin 37 - \mu N_B = 0$$

$$40 - 30.5 - \mu \cdot 39.93 = 0$$

$$\mu \approx 0.25$$